

13 Prove Assignment: Student Chosen Program

Purpose

Prove that you can write a significant Python program that uses real-world data and solves a real world problem. Your program must include multiple functions that you verify are correct with test functions and `pytest`.

Assignment

Finish developing the program that you started in the prove assignment for lesson 11.

Submission

At the end of this lesson, you must submit a description of your work, and your teacher or teaching assistant will grade your work according to the following rubric.

Lesson 13 Rubric

1. Time—50%: Did you spend at least six hours on your Python program or test functions during the current lesson?
2. Description—10%: Is the description of your work for this lesson complete and easily understandable? Your description should include the following:

- a. A list of the function names in your program.
 - b. A list of the test function names in your test code.
 - c. A list of the documentation that you read, the videos that you watched, and the coding experiments that you tried.
 - d. A description or list of the work that you finished on your program.
3. Python program file—25%: Upload your Python program. Your teacher will evaluate it according to these criteria:
- a. Your program is divided into functions and each function performs one task only.
 - b. Your program effectively uses existing Python modules such as math, random, requests, pandas, and tkinter.
 - c. Your program performs a significant real world task.
4. Python test file—15%: Upload your Python test file. Your teacher will evaluate it according to these criteria:
- a. Each testable program function is covered (tested) by one test function. (Some functions, especially main and those that create GUIs are difficult to test and don't need to have a corresponding test function.)
 - b. Each test function completely exercises (tests) its corresponding program function. In other words, the test function calls the program function multiple times with different arguments, including unusual or unexpected values.